these technologies? This article explains these technologies, their applications in different sectors and how to start designing products for them.

AR versus VR

People often use terms like AR, VR and mixed reality (MR) interchangeably to just talk about anything that lies on the real virtual spectrum. It is important to understand their differences to fully grasp the benefits of these individual technologies and know what you need.

AR superimposes media (images, text and the like) onto the real world such that the user is still connected to the surroundings. Snapchat filters with hats or glasses that are highly popular these days utilise AR. Not every AR application uses recognition for sensing though; marker-less AR uses location-detection features like GPS and accelerometer for such purposes as finding nearby places. Cellphones like Asus Zenfone ARES and Google Nexus 6P are compatible with Google's AR platform.

VR, on the other hand, is completely isolated from reality. VR creates the illusion that an image has depth by showing two slightly offset images separately to each eye of the user through a headset or similar device, which creates a simulated 3D digital environment. Frame rate of at least 60fps to process and render images without latency, and field-of-view (FOV) above hundred degrees to support user head or eye movement are some of the necessary considerations while developing an application.

Sound effects also need to be consistent with visuals. For high-end VR applications, sensors placed in the surroundings ensure high accuracy. MR is an amalgamation of the two for optimum results, like in Microsoft Hololens.

Raman Talwar, chief executive officer, Simulanis Solutions, says, "In VR, motion tracking raises the level of experience to a whole new level. With the concept of 6DOF (six degrees of freedom), a user can feel free to move and look around in the virtual world. In AR, the general technology of tracking is based on what is called simultaneous localisation and mapping (SLAM), where the camera orients itself to movement of the user.

"In AR, environmental understanding is based on points cloud sensed from camera and is based on the real world. While in VR, complete environment is generated virtually and nothing is real."

Sweden's McDonald's turns its happy meal box into VR goggles (Credit: www.campaignlive.co.uk)

Virtual expansion so far

Numerous businesses in various industries are already present while others are venturing into AR and VR now. Such companies as Imaginate, Immerse and Simulanis offer customised solutions to enterprises. Hemanth Satyanarayana, chief executive officer, Imaginate, says, "Increased computing power available on handheld devices along with depth-sensing features enable easier generation of AR experiences. Cost of smartglasses has reduced significantly with better performance. The variety of brands available has increased. Businesses are looking for faster design reviews, production cycles and quicker troubleshooting techniques." Some of the prominent applications are discussed next.

Entertainment and gaming. This industry was the earliest adopter of XR and relied on fully-immersive experiences to thrive. Today, consumers can virtually experience live entertainment events, such as concerts, exhibitions and sports events, minus the crowd. For instance, in Denver Museum of Nature and Science, visitors get to see dinosaurs instead of their fossils.

Retail. With the advent of the Internet, e-commerce became popular due to the convenience that buyers got by shopping anytime, anywhere. Businesses are going one step ahead by letting customers try the item before purchase. One such example is Decor Matters that allows customers visualise furniture and decor items before making a purchase.

Education and training. Using XR for education provides students with a more realistic view and improves their understanding. Training imparted to newbies regarding proper utilisation of costly equipment in an industry can not only be complex but mishandling equipment can also cause monetary losses. Moreover, training for professions in fields like military, healthcare, aeronautics and the like often involves dangerous situations that may even be life-threatening.

In many areas today, all three military services (army, navy and air force) are employing AR and VR for vehicle simulation, flight simulation, battlefield simulation, medic training and more at a cheaper cost than most traditional methods.

Remote collaboration. Employees around the world can connect with each other for meetings in a way that makes both sides feel like they are in the same room. Data can also be accessed remotely whenever required.

Real estate. From easing the process of designing homes for architects and interior designers to letting poten-